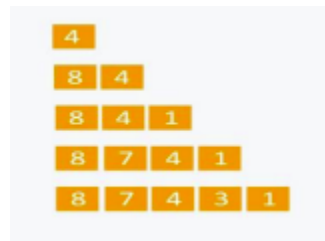


Activity- 1:

In contrast to the standard FIFO implementation of Queue, the LifoQueue uses last-in, first-out ordering (normally associated with a stack data structure). Implement LIFO queue using queue module.

Activity- 2:

We have an array of 5 elements: [4, 8, 1, 7, 3] and we have to insert all the elements in the max-priority queue. First as the priority queue is empty, so 4 will be inserted initially. Now when 8 will be inserted it will move to front as 8 is greater than 4. While inserting 1, as it is the current minimum element in the priority queue, it will remain in the back of priority queue. Now 7 will be inserted between 8 and 4 as 7 is smaller than 8. Now 3 will be inserted before 1 as it is the 2nd minimum element in the priority queue. All the steps are represented in the diagram below:

**Activity- 3:**

Consider a simple priority queue implementation for scheduling the presentations of students based on their roll number. Here roll number decides the priority of the student to present. Since it is a min-heap, roll number 1 is considered to be of the highest priority.

Activity- 4:

Initialize a heap with student marks and their names.

- Print the heap.
- Print the name of students from highest to lowest marks scored by students.